

CSMFAB000000000113-Vol-05

Multi-Vessel Configuration Geometry and Optimization

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1. The n-Vessel Circle: Geometric Framework

n vessels are positioned at the vertices of a regular n-gon inscribed in a circle of radius R on the ocean surface. The launch point is at the center of this circle, at depth d below the surface.

Each vessel holds one Smart Rope connected to the base of the spear, creating a symmetric tensor array.

1.1 Complete Force Analysis

For n vessels at radius R, depth d:

Rope length:

$$L_{\text{rope}} = \sqrt{R^2 + d^2}$$

Rope angle from vertical (alpha):

$$\alpha = \arctan(R/d)$$

Vertical efficiency factor:

$$\eta_v = \cos(\alpha) = d / \sqrt{R^2 + d^2}$$

Lateral cancellation check: The horizontal components from n symmetric ropes cancel completely:

$$\text{Sum}(F_h) = F_{\text{rope}} * \sin(\alpha) * \text{Sum}(\cos(2\pi k/n)) \text{ for } k=0..n-1 = 0 \quad [\text{always, for any } n \geq 2]$$

1.2 Configuration Table: Practical n Values

n	Min Vessel Spacing (m) at R=6m	Rope Angle	Vert Efficiency	Notes
2	12.0 m	11.3°	98.1%	Minimum viable; poor stability
3	10.4 m	11.3°	98.1%	Triangular — simplest practical
4	8.5 m	11.3°	98.1%	Square — quadrupled force
6	6.0 m	11.3°	98.1%	Hexagonal — recommended
8	4.6 m	11.3°	98.1%	Close packing limit at R=6m
10	3.7 m	11.3°	98.1%	Requires careful coordination
12	3.1 m	11.3°	98.1%	High force; tight vessel spacing

Recommended starting configuration: n=6 hexagonal — balance of force multiplication, vessel spacing, coordination complexity, and practical sea state maneuverability.

1.3 Force Scaling with n

Total vertical impulse (from Vol 02):

$$J_{\text{total}} = n * J_{\text{rope}} * \eta_v$$

For $J_{\text{rope}}=1001 \text{ N} \cdot \text{s}$, $\eta_v=0.981$: - $n=3$: $J_{\text{total}} = 2,944 \text{ N} \cdot \text{s} \rightarrow \Delta v = 1,472 \text{ m/s}$ for 2 kg spear - $n=6$: $J_{\text{total}} = 5,888 \text{ N} \cdot \text{s} \rightarrow \Delta v = 2,944 \text{ m/s}$ - $n=9$: $J_{\text{total}} = 8,832 \text{ N} \cdot \text{s} \rightarrow \Delta v = 4,416 \text{ m/s}$ - $n=12$: $J_{\text{total}} = 11,776 \text{ N} \cdot \text{s} \rightarrow \Delta v = 5,888 \text{ m/s}$

Note: These are theoretical maxima. Rope contraction speed limits effective impulse (see Vol 02 correction).

2. Optimal R/d Ratio Study

Varying R while keeping $n=6$ and $d=30\text{m}$:

R (m)	L_rope (m)	alpha	eta_v	Notes
1	30.02	1.9°	99.9%	Vessels nearly over launch point
3	30.15	5.7°	99.5%	Tight but achievable
6	30.60	11.3°	98.1%	Recommended
10	31.62	18.4°	94.9%	Acceptable
20	36.06	33.7°	83.2%	Significant loss
30	42.43	45.0°	70.7%	Poor efficiency
50	58.31	59.0°	51.5%	Not viable

Optimal R = 0.1 to 0.2 * d (99.5% to 98.1% efficiency)

For $d=30 \text{ m}$: **R = 3 to 6 m** is optimal. Practical constraint: vessel length ~10-15 m $\rightarrow$ vessels cannot be closer than ~5 m edge-to-edge At $n=6$, $R=6 \text{ m}$: inter-vessel arc spacing = $2R\sin(\pi/n) = 6 \text{ m}$ — just sufficient for small vessels.

3. Three-Dimensional Vector Analysis

When sea state causes vessel displacement from ideal position:

If vessel k is displaced by error vector (dx_k, dy_k) :

$$dF_{\text{vertical}_k} = -F_{\text{rope}} * (dx_k * R_x + dy_k * R_y) / (L_{\text{rope}}^2)$$

Acceptable position error for < 5% force deviation:

$$\Delta r_{\text{max}} = 0.05 * L_{\text{rope}}^2 / (F_{\text{rope}} * R) = 0.05 * 936 / (50000 * 6) = 0.00016 \text{ m}$$

This is extremely tight (0.16 mm)! In practice, position errors are 1-5 m at sea.

Mitigation: The Smart Rope has elastic compliance that compensates for position errors. The rope acts as a spring-damper — at 10% strain, a 1 m position error in 30 m rope = 3.3% strain variation = 3.3% force variation. **Acceptable.**

4. Sea Anchor Configuration for Vessel Stability

Each vessel must be anchored to maintain position during the brief (0.04 s) maximum-tension event.

Impulse force on each vessel (n=6):

$$F_{\text{vessel_peak}} = F_{\text{rope_peak}} = 50,000 \text{ N (toward center, downward component)}$$

For a 5-tonne vessel: $\text{acceleration} = F/m = 50000/5000 = 10 \text{ m/s}^2 = 1g$ Over $t=0.04 \text{ s}$: $\text{displacement} = 0.5 * 10 * 0.04^2 = 0.008 \text{ m} = 8 \text{ mm}$

8 mm vessel displacement during launch is acceptable — no anchoring required for short launches. For repeated launches, a BFRP anchor chain (non-conductive) is recommended.

Citations (Vol 05)

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